

Definition. A *group homomorphism* ϕ from a group G to a group $\overline{G}$ is

Definition. The *kernel* of a homomorphism ϕ , $\text{Ker}\phi$ is:

Exercise 1. For each of the following mappings, determine (1) whether ϕ is an isomorphism, (2) whether ϕ is a homomorphism, (3) the kernel of ϕ .

- Let $\phi : \mathbb{Z} \rightarrow \mathbb{Z}_2$ such that $\phi(x) = \begin{cases} 1 & \text{if } x \text{ is odd} \\ 0 & \text{if } x \text{ is even} \end{cases}$

- Let $G = \{ax + b : a, b \in \mathbb{R}\}$ (the set of degree 0 and 1 polynomials over $\mathbb{R}$) and $\phi : G \rightarrow G$. Let $\phi(ax + b) = \frac{d}{dx}(ax + b)$.

- Let $\phi : \mathbb{R} \rightarrow \mathbb{R}$. Let $\phi(x) = x^2$.

Exercise 2. Define a homomorphism from $\mathbb{Z}$ to $\mathbb{Z}_5$.

Theorem.

Theorem 10.1: Properties of Homomorphisms

Let $\phi : G \rightarrow \overline{G}$ be a group homomorphism. Let G have identity e and $\overline{G}$ have identity $\bar{e}$. Then

1. $\phi(e) = \bar{e}$.
2. $\phi(g^n) = (\phi(g))^n$ for all $n \in \mathbb{Z}$.
3. If $|g|$ is finite, then $|\phi(g)|$ divides $|g|$.
4. $\text{Ker}\phi$ is a subgroup of G .
5. $\phi(a) = \phi(b)$ iff $a\text{Ker}\phi = b\text{Ker}\phi$.
6. If $\phi(g) = g'$, then $\phi^{-1}(g') = \{x \in G \mid \phi(x) = g'\} = g\text{Ker}\phi$.

Theorem 10.2: Properties of Subgroups Under Homomorphisms

Let $\phi : G \rightarrow \overline{G}$ be a group homomorphism, and let $H \leq G$. Let G have identity e and $\overline{G}$ have identity $\bar{e}$. Then

1. $\phi(H) = \{\phi(h) \mid h \in H\}$ is a subgroup of $\overline{G}$.
2. If H is cyclic, then $\phi(H)$ is cyclic.
3. If H is Abelian, then $\phi(H)$ is Abelian.
4. If $H \triangleleft G$, then $\phi(H) \triangleleft \overline{G}$.
5. If $|\text{Ker}\phi| = n$, then ϕ is an n -to-1 mapping from G onto $\phi(G)$.
6. If $|H| = n$, then $|\phi(H)|$ divides n .
7. If $\overline{K} \leq \overline{G}$, then $\phi^{-1}(\overline{K}) = \{k \in G \mid \phi(k) \in \overline{K}\} \leq G$.
8. If $\overline{K} \triangleleft \overline{G}$, then $\phi^{-1}(\overline{K}) = \{k \in G \mid \phi(k) \in \overline{K}\} \triangleleft G$.
9. If ϕ is onto and $\text{Ker}\phi = \{e\}$, then ϕ is an isomorphism from G to $\overline{G}$.

Exercise 3. Let G be a group and H be any normal subgroup of G . Define $\phi : G \rightarrow G/H$ by $\phi(g) = gH$. Prove ϕ is a homomorphism and find the kernel.

Theorem (First Isomorphism Theorem).

Theorem.

ADDITIONAL EXERCISES

1. Suppose that there is a homomorphism ϕ from $\mathbb{Z} \oplus \mathbb{Z}$ to a group G such that $\phi((3, 2)) = a$ and $\phi((2, 1)) = b$. Determine $\phi((4, 4))$ in terms of a and b , assuming the operation is addition.
2. Can there be a homomorphism from $\mathbb{Z}_4 \oplus \mathbb{Z}_4$ onto $\mathbb{Z}_8$? Explain.
3. Can there be a homomorphism from $\mathbb{Z}_{16}$ onto $\mathbb{Z}_2 \oplus \mathbb{Z}_2$? Explain.
4. How many homomorphisms are there from $\mathbb{Z}_{20}$ onto $\mathbb{Z}_8$? What about to $\mathbb{Z}_8$.
5. Consider a homomorphism ϕ from $U(40)$ to $U(40)$ with kernel $\{1, 9, 17, 33\}$ and $\phi(11) = 11$. Find all elements that map to 11 under ϕ .

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6. Determine all homomorphic images of D_4 up to isomorphism.

 7. Let p be a prime. Determine the number of homomorphisms from $\mathbb{Z}_p \oplus \mathbb{Z}_p$ into $\mathbb{Z}_p$.

 8. Suppose that ϕ is a homomorphism from a group G onto $\mathbb{Z}_6 \oplus \mathbb{Z}_2$ and that $\text{Ker}\phi$ has order 5. Explain why G must have normal subgroups of orders 5, 10, 15, 20, 30, and 60.

 9. Let G be an Abelian group. Let $K = \{a \in G : a^2 = 1\}$ and let $H = \{x^2 : x \in G\}$. Use the First Isomorphism Theorem to prove that G/K is isomorphic to H .

10. Find a homomorphism ϕ from $\mathbb{Z}$ to some group such that $\mathbb{Z}/\text{Ker}\phi \approx \mathbb{Z}_6$.
11. Let H be a subgroup of a group G and let K be a normal subgroup of G .
- (a) Show that the mapping from H to HK/K given by $\phi(h) = hK$ is a homomorphism onto HK/K .
- (b) If H and K are finite, explain why it follows that $|HK| = |H||K|/|H \cap K|$.